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## Do medications increase the efficacy of digital interventions for smoking cessation? Secondary results from the iCanQuit randomized trial

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### Abstract

**Background and aims** —iCanQuit is a smartphone app proven efficacious for smoking cessation in a Phase III randomized controlled trial (RCT). This study aimed to measure whether medications approved by the United States Food and Drug Administration (FDA) for smoking cessation would further enhance the efficacy of iCanQuit, relative to its parent trial comparator—the National Cancer Institute’s (NCI’s) QuitGuide app.

**Design** —Secondary analysis of the entire parent trial sample of a 2-group (iCanQuit and QuitGuide), stratified, doubled-blind RCT.

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Credit authorship contribution statement

Jonathan J. Bricker and Kristin Mull conceptualized the study. Jonathan J. Bricker led manuscript writing. Kristin Mull led and conducted data analysis. All authors assisted in manuscript writing and provided critical review. All authors have read and agreed to the published versions of the manuscript.

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Study protocol can be found in the supplement content of the primary outcome published paper (Jonathan JB et al. JAMA Intern Med. 2020;180(11): 1472–1480).

**Setting** —United States of America.

**Participants** —Participants who reported using an FDA-approved cessation medication on their own (n = 619) and those who reported no use of cessation medications (n = 1469).

**Interventions** —Participants were randomized to receive iCanQuit app or NCI's QuitGuide app.

**Measurements** —Use of FDA-approved medications was measured at 3 months post-randomization. Smoking cessation outcomes were measured at 3, 6 and 12 months. The primary outcome was 12-month self-reported 30-day point prevalence abstinence (PPA).

**Findings** —The data retention rate at the 12-month follow-up was 94.0%. Participants were 38.5 years old, 71.0% female, 36.6% minority race/ethnicity, 40.6% high school or less education, residing in all 50 US States, and smoking 19.2 cigarettes/day. The 29.6% of all participants who used medications were more likely to choose nicotine replacement therapy (NRT; 78.8%) than other cessation medications (i.e., varenicline or bupropion; 18.3%, 10.5%, respectively) and use did not differ by app treatment assignment (all  $p > .05$ ). There was a significant ( $p = .049$ ) interaction between medication use and app treatment assignment on PPA. Specifically, 12-month quit rates were 34% for iCanQuit vs. 20% for QuitGuide (odds ratio [OR] = 2.36; 95% confidence interval [CI]: 1.59, 3.49) among participants reporting any medication use, whereas among participants reporting no medication use, quit rates were 28% for iCanQuit vs. 22% for QuitGuide (OR = 1.41; 95% CI: 1.09, 1.82). Results were stronger for those using only NRT: 40% quit rates for iCanQuit vs. 18% quit rates for QuitGuide (OR = 3.57; 95% CI: 2.20, 5.79).

**Conclusions** —The iCanQuit smartphone app for smoking cessation was more efficacious than the QuitGuide smartphone app, regardless of whether participants used medications to aid cessation. Smoking cessation medications, especially nicotine replacement therapy, might enhance the efficacy of the iCanQuit app.

### Keywords

ACT; Acceptance & Commitment Therapy; digital interventions; iCanQuit; smartphone applications; smoking cessation; pharmacotherapy; nicotine replacement therapy

## INTRODUCTION

Cigarette smoking is a leading cause of early death and disability, accounting for more than 1 in 10 deaths worldwide [1]. Barriers to accessing smoking cessation treatments include limited access to evidence-based treatments, low reimbursement for clinicians, and low demand for in-person treatment [2]. Smartphone applications (apps) for smoking cessation have been addressing access barriers by serving as digital therapeutics with high potential for population-level reach [3]. There are now over 500 English-language smoking cessation applications, which have been downloaded over 33 million times [4]. In the United States (US), the reach of smoking cessation applications has been aided by the fact that, as of 2021, 85% of all adults owned smartphones—up from 35% in 2011 [5].

We published a large (N = 2415) randomized controlled trial (RCT) to determine the efficacy of a smartphone app for smoking cessation (iCanQuit) based on Acceptance and

Commitment Therapy (ACT), compared with an NCI smoking cessation app (QuitGuide) based on the US Clinical Practice Guidelines (USCPG) [6]. Previous studies have demonstrated that ACT-based interventions promote smoking cessation by teaching adults who smoke to accept their cravings to smoke and let these cravings pass without smoking [7–16]. Compared to standard approaches for tobacco cessation, such as the USCPG that teaches avoidance of cravings, ACT-based processes teach people to observe and accept emotions and cravings that cue smoking. At the 12-month follow-up, iCanQuit was 1.5 times more efficacious than QuitGuide for smoking cessation. Moreover, results from this study showed that the effect of the intervention on smoking cessation was mediated through greater acceptance of cues to smoke [8]. iCanQuit is the first app proven efficacious for smoking cessation in a full-scale RCT with long term follow-up.

While iCanQuit is efficacious for smoking cessation, it has to date only been tested when offered as a standalone behavioral intervention. Yet the combination of behavioral *and* pharmacotherapy is recommended by USCPG as the most effective approach to quitting smoking, as this approach targets both behavioral skills for enhancing motivation and coping with cravings and the physiological underpinnings of cravings and withdrawal [17]. While it is well known that the combination of behavioral intervention with pharmacotherapy generally produces higher quit rates than behavioral intervention alone [18–22], these conclusions are based on the large body of research that tested behavioral interventions delivered in-person or via telephone by trained smoking cessation specialists or medical personnel. Such an approach to combination treatment has high financial and time costs to deliver (e.g., costs to patients, low provider reimbursement, time to train and deliver behavioral treatment) and is thus challenging to disseminate [23, 24].

Whether adding pharmacotherapy to self-help interventions, such as digital apps, mobile texting, or websites, would further improve smoking cessation rates is largely unknown [3, 22, 25–27]. In a pilot study of non-treatment seeking primary care patients who smoke, Kruse et al. [28] found that the text messaging group ( $n = 39$ ) had similar 3-month quit rates as the text messaging plus 2-week supply nicotine replacement therapy (NRT) group ( $n = 39$ ), though these results were limited by small sample sizes and short follow-up length. And the corollary to this question, whether digital interventions enhance the efficacy of pharmacotherapy for smoking cessation, is largely unexplored. Specifically, Vidrine et al. [29] found that a text messaging intervention added to a mailing of a 10-week supply of nicotine patches was not more effective for smoking cessation than nicotine patches alone in a community sample of low-income adult smokers. By contrast, Carrasco-Hernandez et al. [30] found that tailored text messaging plus face-to-face behavioral therapy plus varenicline or bupropion was more efficacious for smoking cessation than face-to-face behavioral therapy plus varenicline or bupropion in sample of smokers in an outpatient clinic in Seville, Spain. Both studies were limited by high outcome data attrition (range: 27% to 60%). Taking into account this small body of research, novel scientific knowledge would be gained from examining the question of whether pharmacotherapy improves the efficacy of existing digital interventions shown efficacious for smoking cessation. Given that digital interventions are becoming the most common self-help intervention for smoking [3, 31, 32], understanding whether medications improve the efficacy of smoking cessation apps is a timely question.

The potential public health significance of examining the efficacy of the iCanQuit app combined with pharmacotherapy is high. First, digital interventions like iCanQuit are less costly than in-person or telephone-based interventions and are broadly disseminable self-help interventions [33–35]. Second, mailed pharmacotherapy, especially NRT patch, gum, and lozenge (or their combination), are modest-cost interventions that are also broadly disseminable, and can be safely provided with minimal medical oversight [36–41]. Third, as both iCanQuit and pharmacotherapy are each proven efficacious for smoking cessation, their combination has the potential to achieve even higher quit rates. Finally, since public health impact is a product of both population-level reach and efficacy [42], the combination of both iCanQuit and pharmacotherapy could have high impact.

Given the novelty and importance of examining this combination, the overall aim of this study is to examine whether the use of FDA-approved medications for smoking cessation would further enhance the efficacy of iCanQuit. We first explored whether there was an interaction between any FDA-approved cessation pharmacotherapy use (i.e., NRT, varenicline, bupropion) and app treatment assignment (iCanQuit vs. QuitGuide) on the 12-month 30-day PPA main outcome of the iCanQuit parent trial. iCanQuit teaches individuals to use pharmacotherapy to manage smoking triggers in the short run while learning skills for accepting triggers in the long run. By contrast, QuitGuide presents pharmacotherapy as a separate treatment that is used alongside or in addition to the behavioral skills. Thus, it may not be as clear in QuitGuide how a participant can use the two treatments together in a conceptually coherent manner. Given the differences between the two apps in how they address pharmacotherapy and behavioral treatment, an interaction test was used to directly determine whether the effect pharmacotherapy use on smoking cessation differs between treatment arms [43]. We further hypothesized that iCanQuit participants who used pharmacotherapy would be more likely to quit smoking than those who used iCanQuit without pharmacotherapy. Finally, to explore the potential value of NRT as a highly disseminable adjunct to digital interventions, all analyses were repeated by comparing participants in each arm who reported only using NRT (i.e., patch, gum, lozenge, and/or inhaler) vs. no pharmacotherapy at all.

## METHODS

### Overview

Full details on the iCanQuit trial methods have been reported [6]. Briefly, eligibility criteria included age 18 and older, daily cigarette smoking, smartphone access, and wanting to quit smoking within the next 30 days. Exclusion criteria included being unable to read English, using any pharmacotherapy, behavioral or other therapies for smoking cessation, having used either iCanQuit or QuitGuide in the past, or having a household member already enrolled in the study. Participants were recruited and screened for eligibility online. The parent trial was a racially/ethnically diverse sample of 2415 adults from all 50 US states who were randomized 1:1 to receive an ACT-based smartphone application (iCanQuit) or a USCPG-based smartphone application (QuitGuide) for 12 months [6]. Randomization by permuted blocks of size 2, 4, and 6 was stratified by daily smoking frequency ( $\leq 20$  vs  $\geq 21$  cigarettes/day), minority race/ethnicity, education level ( $\leq$  high school

vs  $\geq$ some college), and positive screening for depression (Center for Epidemiological Studies-Depression (CESD)-20 scale score  $\leq 15$  vs  $\geq 16$ ). Study procedures were approved by the Fred Hutchinson Cancer Center Institutional Review Board. All participants were compensated to up to \$105 for completing study data collection.

### Study population, recruitment, and enrollment

The study population for the current study was defined as all parent trial participants who reported using an FDA-approved cessation medication within the first three months following randomization into the trial ( $n = 619$ ) and those who reported no use of cessation medications during this period ( $n = 1469$ ), and thus the total sample size was 2088 (86.5% of parent trial sample). These participants were recruited via Facebook ads (1694/2088, 81.1%), a survey sampling company (295/2088, 14.1%), word of mouth (56/2088, 2.7%), and websites or search engine results (43/2088, 2.1%). Participants were enrolled between May 2017 and September 2018. Participants were given unique access to their assigned application and were sent unique links to study surveys online at the 3-, 6-, and 12-month follow-ups. Follow-up data collection was between August 2017 and December 2019.

### Interventions

**iCanQuit**—Table 1 shows the major similarities and differences between iCanQuit and QuitGuide, including information on the FDA-approved medications for quitting smoking. Participants randomized to the iCanQuit arm received access to download the iCanQuit smartphone app [6]. iCanQuit teaches ACT skills for coping with smoking urges, staying motivated, and preventing relapse. The program provides information on the FDA-approved medications for quitting smoking. iCanQuit is self-paced, and content is unlocked in a sequential manner. The app teaches skills to accept physical sensations, emotions, and thoughts that trigger smoking via distancing from thoughts about smoking, mindfulness, and perspective taking. For the first four levels, exercises are unlocked immediately after the prior exercise is complete. The first four levels contain content and exercises designed to prepare users for their chosen quit day. For the last four levels, the next level does not unlock until users record seven consecutive smoke-free days. Levels two to four contain 26 exercises teaching skills to accept cravings, emotions, and thoughts that trigger smoking. The last four levels contain content and exercises designed to help the user stay smoke-free after their quit date. These levels contain 25 exercises that focus on building smoke-free life activities and coping with withdrawal symptoms, slips, depression, and potential weight gain. If the user lapses (e.g., records having smoked a cigarette), the program encourages (but does not require) them to set a new quit date and return to the first four levels for preparation.

**QuitGuide**—Participants randomized to the QuitGuide arm received access to download the National Cancer Institute's QuitGuide smartphone app [6]. QuitGuide focuses on increasing motivation to quit by using reason and logic and providing information on the health consequences of smoking (i.e., expectancies). The app helps users develop a quit plan, identify smoking behaviors, triggers, and reasons for being smoke-free, and to identify sources of social support for quitting. It teaches skills for avoiding situations that lead to cravings to smoke, staying smoke-free, and coping with slips. Main components are as

follows: (1) “Thinking about quitting”, which focuses on motivations to quit by encouraging users to think of reasons for quitting and providing information on the general health consequences of smoking and quitting; (2) “Preparing to Quit”, which helps users develop a quit plan, identify smoking behaviors, triggers, and reasons for being smoke-free, identify social support for quitting, and provides information on the FDA-approved medications for quitting smoking; (3) “Quitting”, which teaches skills for avoiding cravings to smoke, such as finding replacement behaviors (e.g., chewing on carrot sticks) and staying busy; and (4) “Staying Quit”, which presents tips, motivations, and actions to stay smoke-free and skills for coping with slips via fighting cravings and trying to be positive. No quit smoking pharmacotherapies, coaching, or any other intervention was provided in either arm.

## Study Measures

**Baseline characteristics**—Baseline data included age, gender, race/ethnicity, education, employment, income, marital status, sexual orientation, and zip codes. Study participants completed positive screening tools to assess mental health, including depression [44], panic [45], and post-traumatic stress (PTSD) disorders [46]. Alcohol consumption was assessed via the Quick Drinking Screen [47]. Smoking behavior variables included nicotine dependence (measured by Fagerström Test for Cigarette Dependence) [48], number of cigarettes smoked per day, years of smoking, use of e-cigarettes, quit attempts, confidence to quit smoking, and relationships with other people who smoke.

**FDA-approved medications**—At the three-month follow-up survey, use of FDA-approved medications was assessed with the following question: “Since the date you joined the study, did you ever use any of the following nicotine replacement therapies or medications to help you quit smoking?” Participants were defined as using any FDA-approved cessation pharmacotherapy if they selected nicotine gum, nicotine patch, varenicline, bupropion, and/or entered nicotine lozenge or inhaler, either alone or in combination with other FDA-approved medications. Participants were defined as using *only* NRT if they selected nicotine gum, nicotine patch, and/or entered nicotine lozenge or inhaler, either alone or in combination thereof.

**Smoking cessation outcomes**—Consistent with the iCanQuit parent trial [6], primary analysis was performed on a complete-case basis, with intent-to-treat missing-as-smoking and multiple imputation sensitivity analyses. Complete-case analysis is appropriate in studies with low attrition and when baseline covariates related to the outcome are included in the analysis [49]. The primary smoking cessation outcome was self-reported 30-day point-prevalence abstinence (PPA) at 12 months. Secondary smoking cessation outcomes were: 7-day PPA, missing-as-smoking imputation, multiple imputation, prolonged abstinence defined as no smoking at all in the 9-month period of 3 to 12 months post-randomization, and cessation of all nicotine and tobacco products (e-cigarettes or vaping, chewing tobacco, snus, hookahs, cigars, cigarillos, tobacco pipes, and cloves/kreteks) at 12 months.



## Statistical Analysis

Baseline characteristics were compared between groups by reported use of FDA-approved pharmacotherapy at 3 months using Fisher's exact tests for categorical variables and t-tests for continuous variables. Participant zip codes were characterized as urban or rural by linking them to Rural-Urban Commuting Area (RUCA) codes using the R library 'zipcode' [50, 51]. RUCA codes of 1–3 were considered urban, while codes 4–10 were considered rural [52]. To determine whether use of pharmacotherapy by 3 months differentially impacted the effect of app treatment arm on the primary 12-month cessation outcome, we used a logistic regression model with a pharmacotherapy use by treatment interaction term. Following on the results of the interaction analysis, logistic regression models were used to estimate the app treatment effect for participants who used pharmacotherapy and those who did not use pharmacotherapy, and to estimate the effect of pharmacotherapy use for participants in the iCanQuit arm. For multiple imputation of missing 12-month smoking status, we used multivariate imputation by chained equations in the R package 'mice' [53] to create ten complete data sets and pool model results. Similar interaction analyses and logistic regression models were used to investigate the association between use of NRT only (vs. no pharmacotherapy) and smoking cessation. In all models, we adjusted for the four baseline factors used in stratified randomization: minority race or ethnicity, high school or less education, heavy smoking (i.e., >20 cigs/day), and positive depression screen (i.e., CESD-20  $\geq 16$ ) [54]. We also adjusted for baseline participant characteristics that were significantly associated with the outcome, in order to avoid obtaining incorrect confidence intervals [55, 56]. Unadjusted analyses were explored for cessation outcomes in the iCanQuit arm. All statistical tests were 2-sided, with  $\alpha = 0.05$ , and analyses were completed using R version 4.2.2. [57].

## RESULTS

### Participant pharmacotherapy use, and retention rates

At the 3-month follow-up timepoint, 29.6% (619/2088) of participants indicated that they had used FDA-approved pharmacotherapy since joining the study, while 70.4% (1469/2088) had not. Use of pharmacotherapy did not differ by treatment arm: 29% (300/1041) of iCanQuit participants reported using any pharmacotherapy vs. 30% (319/1047) of QuitGuide participants ( $p = 0.468$ ). As shown in Table 2, those using pharmacotherapy by the three-month follow-up were slightly more likely to be non-Hispanic White race and have higher income but were more likely to be disabled. In addition, pharmacotherapy users had higher baseline nicotine dependence and were more likely to have smoked  $\geq 10$  years. Seventy-nine percent of all pharmacotherapy users reported using some form of NRT (nicotine patch, 302/619, 48.8%; nicotine gum, 260/619, 42.0%; lozenges or inhalers, 19/619, 3.1%), while thirty percent reported using other FDA-approved medications (varenicline, 113/619 18.3%; bupropion, 81/619 10.5%). Note that percentages do not sum to 100% because participants were able to select more than one pharmacotherapy type. A smaller number of users (55/619, 8.9%) chose a combination of NRT and medication.

The 12-month follow-up outcome data retention rate was 94% (1962/2088) overall, and this rate did not differ by treatment arm (93% (971/1041) for iCanQuit; 95% (991/1047) for

QuitGuide,  $p = .174$ ) or by users vs. non-users of medications (94% (583/619) for users of medications; 94% (1379/1469) for non-users of medications,  $p = .892$ ).

### Smoking cessation

Using any type of FDA-approved pharmacotherapy by 3-months post-randomization differentially impacted the effect of treatment on 12-month smoking cessation (Wald  $p = .033$ ; Figure 1). Among participants who did not use pharmacotherapy, the primary outcome 12-month follow-up 30-day PPA rates were 28% for iCanQuit vs. 22% for QuitGuide ( $p = .009$ ). Among participants who used pharmacotherapy, the 12-month follow-up 30-day PPA rates were 34% for iCanQuit vs. 20% for QuitGuide ( $p < .001$ ). Within the iCanQuit arm, evidence suggested that using pharmacotherapy was associated with 1.37 times higher odds of quitting smoking by 12 months ( $p = .051$ ; Table 3). The trend was similar for other secondary cessation outcomes, but none were statistically significant (Table 3). Supplementary Table 1 shows results of logistic regression models in the iCanQuit arm that are unadjusted for covariates, in order to demonstrate the stability of results across covariate-adjusted and unadjusted analyses.

The enhancing effect of medication was strong for the use of NRT as the only form of pharmacotherapy. There was a significant interaction effect (Wald  $p = .001$ ) between use of only NRT and app treatment assignment on the main outcome of 12-month follow-up 30-day PPA (Figure 2). Specifically, among participants who only used NRT, 30-day PPA rates at 12 months were 40% for iCanQuit vs. 18% for QuitGuide ( $p < .001$ ; Table 4). iCanQuit participants who used NRT only had 1.78 times higher odds of quitting smoking at the 12-month follow-up as compared to those who did not use any pharmacotherapy (40% vs. 28%;  $p = .002$ ; Table 4).

For nearly all secondary outcomes within the iCanQuit treatment arm, there was evidence that the use of NRT (i.e., as the only form of pharmacotherapy for smoking cessation) was associated with higher odds of cessation (Table 4). The use of NRT only *or* in combination with other pharmacotherapy was also associated with cessation in the iCanQuit arm, though these results were not as strong as the results for use of NRT *alone* (Supplementary Table 2).

### Motivation to quit

We explored whether motivation to quit was a potential confound on the cessation results for medication vs non-medication users, using two indicators of motivation: self-reported confidence in quitting at baseline and level of app engagement. Table 2 showed no significant differences in confidence in quitting between medication users and non-users. There were no differences in app engagement (as measured by the standard of mean number of app openings) between medication users and non-users in the iCanQuit arm (M(SD): 41.1 (82.5) vs. 43.3 (98.5);  $p = .446$ ). By contrast, the mean number of app openings was higher for medication users than non-users in the *QuitGuide* arm (M(SD): 16.4 (93.4) vs. 8.5 (12.8);  $p = .013$ ).



## DISCUSSION

The overall aim of this study was to examine whether FDA-approved medications for smoking cessation would further enhance the efficacy of the iCanQuit app intervention for smoking cessation. Results supported all the study hypotheses. First, the significant interaction results showed improved cessation outcomes among iCanQuit vs. QuitGuide participants who started using medication by 3-months following randomization. Among those using any cessation pharmacotherapy, iCanQuit participants were more than twice as likely to quit smoking than QuitGuide participants. Within the iCanQuit arm, using any pharmacotherapy was associated with nearly one and half times higher odds of quitting smoking than not using pharmacotherapy. Finally, among those using only NRT, iCanQuit participants had over three and half higher odds of quitting smoking than QuitGuide participants. Within the iCanQuit arm, using NRT was associated with 1.78 times higher odds of quitting smoking compared to not using NRT. Notably, both the missing-as-smoking imputation and multiple imputation sensitivity analyses provided evidence for the stability of primary outcome results.

There was no evidence that motivation to quit was a confounding factor for the observed results that medication users had higher cessation rates than non-users in the *iCanQuit* arm. Higher app engagement among medication users (vs. non-users) in the *QuitGuide* arm did not translate into higher cessation rates. Overall, there was no additive effect of medications for a standard app that follows the US Clinical Practice Guidelines (i.e., *QuitGuide*) but there was an additive effect of medications, which was particularly strong for NRT, for an app that follows Acceptance and Commitment Therapy (i.e., *iCanQuit*). The significance of this finding is that medications may add value to a digital behavioral intervention, but that added value seems to depend on which digital intervention is utilized. More broadly, the set of results are more nuanced than the general view that medication added to behavioral interventions is more effective than behavioral interventions alone—as that effect may depend on the behavioral intervention [18–22].

The results suggest a synergistic effect of pharmacotherapy for iCanQuit. The app follows the ACT approach to behavior change, which teaches participants to “change what they can and accept what they cannot” [58, 59]. Within this framework, pharmacotherapy is considered an approach for changing what can be changed—in this case using a biological agent to change the intensity and frequency of cravings and withdrawal symptoms that trigger smoking. Complementing this approach, ACT’s acceptance skills teach participants how to be aware, open to, and willing to experience any triggers for smoking that are not being managed or controlled by medications [60, 61]. The overall therapeutic message is to use pharmacotherapy to manage triggers in the short term while they learn skills for accepting them in the longer term. This coherent blending of ACT’s specific behavioral skills with pharmacotherapy likely explains why the combination of pharmacotherapy with iCanQuit was more effective than iCanQuit alone. While using acceptance skills alone might have been helpful, as suggested by the higher quit rates for iCanQuit vs. *QuitGuide* among non-medication users, the results suggest that using this complementary approach of both acceptance (i.e., ACT acceptance training) and change (i.e., pharmacotherapy) was the more effective method.

In contrast, there was no enhancing effect of medications for the QuitGuide treatment arm. While the combination of behavioral intervention with pharmacotherapy generally produces higher quit rates than behavioral intervention alone [18–21], we think that a challenge to making this combination work is in how the role of pharmacotherapy is framed within the context of digital interventions like smartphone apps. Like in many apps that follow the USCPG, QuitGuide presents pharmacotherapy as a separate treatment that is used alongside or in addition to the behavioral skills. Thus, it is not clear how a participant can use the two treatments together and in a conceptually coherent manner. By contrast, when this combination is delivered by a clinician or a quitline coach, there is ample opportunity for the clinician to clarify their dual roles and to offer a compelling rationale to the participant. While future research is needed to understand why QuitGuide with medications did not improve cessation outcomes, our interpretation is that participants might not have understood the complementary functions of each treatment, and thus experienced no added benefit to the combination. And while there were contrasts in how the two apps addressed responding to common medication side effects, the level of these contrasts is unlikely to account for the magnitude of the observed differences in cessation effects between the two apps.

NRT alone appears to have strong enhancing effects for iCanQuit, with an observed 30-day PPA quit rate of 40% at the 12-month follow-up. While results were similar for NRT in combination with other pharmacotherapies, the strong results for NRT alone have high practical significance. NRT is a modest-cost intervention that is broadly disseminable and can be safely provided with minimal medical oversight [36–41]. Indeed, a number of clinical trials have shown that simply mailing NRT has been proven effective for smoking cessation [62, 63]. Building on the current study's results, needed next is a randomized clinical trial to determine whether providing NRT along with iCanQuit is more efficacious for smoking cessation than iCanQuit alone.

There are several important limitations to the current study. First, while participants were randomly assigned to the iCanQuit vs. QuitGuide arms, they were not randomly assigned to the provision (vs. no provision) of pharmacotherapy. Thus, it is possible that participants who elected to use pharmacotherapy on their own may have differed at baseline from those who did not use pharmacotherapy on their own. Such baseline differences, while modest (e.g., users were mean age 39.8 years vs. non-users were 37.9 years), were adjusted for in the analysis and this adjustment did not change the overall interpretation of the results. But since we cannot exclude the possibility that the results could be influenced by unmeasured confounders, the results should be viewed as exploratory. Second, the dosages of, durations of, combinations of, and adherence to pharmacotherapy use were not examined, and the use of pharmacotherapy was only self-reported, and thus future research should elucidate the effect of these factors on the precision and size of cessation effects. A future question worth exploring in a trial that does collect compliance data is whether the stronger effects of iCanQuit are due to higher compliance to medication use. Third, smoking status was not biochemically verified. As the fraction of NRT use was higher than for other forms of pharmacotherapy, there might have been more power to detect effects of NRT than other forms of pharmacotherapy. The self-reported outcome was prespecified based on methodological problems with remote biochemical verification [64,

65]. Although previous studies have demonstrated strong agreement between self-reported and biochemically verified smoking status [66, 67], others showed evidence of significant discordance [68, 69]. Therefore, the external validity of the self-reported smoking status in this trial is not known. However, given the double blinding of the trial, we see no compelling reason why the false reporting rate would be higher in one arm versus the other. Finally, several secondary cessation outcomes were non-significant for the overall pharmacotherapy analysis and were non-significant for the prolonged abstinence secondary outcome for both the overall pharmacotherapy and NRT analyses. While prolonged abstinence is a stringent outcome measure [70, 71], the parent trial was not designed or powered to test for prolonged abstinence [6]. The strongest and most consistent result across the cessation outcomes was for the enhancing effect of NRT.

The study has several key strengths. The results of this study are highly novel, as to date we are aware of only one prior study examining whether adding pharmacotherapy to digital interventions (i.e., apps, texting, websites) would further improve smoking cessation rates. In this pilot study of 78 non-treatment seeking primary care patients who smoke, Kruse et al. [28] found that the text messaging group (n = 39) had similar 3-month quit rates as the text messaging plus 2-week supply nicotine replacement therapy (NRT) group (n = 39). The current study had a large sample size (N = 2088), 12-month follow-up, and at 94% retention rate. The current study's very high retention rate that did not differ by study arm or medication use status results reduces the potential for bias in the missing equals smoking analysis. The study was successful in recruiting a geographically diverse sample, thereby demonstrating potential for high reach, generalization of the results, and broad dissemination.

## Conclusions

In conclusion, cessation medications might enhance the efficacy of iCanQuit for smoking cessation. iCanQuit was more efficacious than QuitGuide, regardless of whether participants used medications to aid cessation. Given the strong effects for NRT, participant preference for NRT, and the efficiencies of broadly disseminating NRT, a full-scale randomized trial testing whether NRT enhances the efficacy of iCanQuit is needed.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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## References:

1. Tobacco Collaborators GBD. Smoking prevalence and attributable disease burden in 195 countries and territories, 1990–2015: a systematic analysis from the Global Burden of Disease Study 2015. *Lancet*. 2017;389(10082):1885–906.PMC5439023. 10.1016/S0140-6736(17)30819-X. [PubMed: 28390697]
2. Husten CG. A Call for ACTION: Increasing access to tobacco-use treatment in our nation. *Am J Prev Med*. 2010;38(3 Suppl):S414–710.10.16/j.amepre.2009.12.006. [PubMed: 20176316]
3. Whittaker R, McRobbie H, Bullen C, Rodgers A, Gu Y, Dobson R. Mobile phone text messaging and app-based interventions for smoking cessation. *Cochrane Database Syst Rev*. 2019;10:CD006611.PMC6804292. 10.1002/14651858.CD006611.pub5.
4. Nelson R. Personal Communication: [SensorTower.com](https://www.sensortower.com). In: Bricker JB, editor.
5. Mobile Fact Sheet [Internet]. 2021 [cited June 25, 2021]. Available from: <https://www.pewresearch.org/internet/fact-sheet/mobile/>.
6. Bricker JB, Watson NL, Mull KE, Sullivan BM, Heffner JL. Efficacy of Smartphone Applications for Smoking Cessation: A Randomized Clinical Trial. *JAMA Intern Med*. 2020;180(11):1472–80.PMC7506605. 10.1001/jamainternmed.2020.4055. [PubMed: 32955554]
7. Bricker JB, Bush T, Zbikowski SM, Mercer LD, Heffner JL. Randomized trial of telephone-delivered acceptance and commitment therapy versus cognitive behavioral therapy for smoking cessation: a pilot study. *Nicotine Tob Res*. 2014;16(11):1446–54.PMC4200023. 10.1093/ntr/ntu102. [PubMed: 24935757]
8. Bricker JB, Levin M, Lappalainen R, Mull K, Sullivan B, Santiago-Torres M. Mechanisms of Smartphone Apps for Cigarette Smoking Cessation: Results of a Serial Mediation Model From the iCanQuit Randomized Trial. *JMIR Mhealth Uhealth*. 2021;9(11):e32847.PMC8663588. 10.2196/32847.
9. Karekla M, Savvides SN. Smoking cessation avatar-led Acceptance and Commitment Therapy digital intervention: feasibility and acceptability in young adults. *Transl Behav Med*. 2021;11(1):198–205.PMC8505109. 10.1093/tbm/ibz128. [PubMed: 31628476]
10. Mak YW, Leung DYP, Loke AY. Effectiveness of an individual acceptance and commitment therapy for smoking cessation, delivered face-to-face and by telephone to adults recruited in primary health care settings: a randomized controlled trial. *BMC Public Health*. 2020;20(1):1719.PMC7667813. 10.1186/s12889-020-09820-0. [PubMed: 33198700]
11. Mak YW, Loke AY, Leung DYP. Acceptance and Commitment Therapy versus Social Support for Smoking Cessation for People with Schizophrenia: A Randomised Controlled Trial. *J Clin Med*. 2021;10(19).PMC8509331. 10.3390/jcm10194304.
12. McCallion EA, Zvolensky MJ. Acceptance and Commitment Therapy (ACT) for smoking cessation: a synthesis. *Curr Opin Psychol*. 2015;2(:):47–51
13. Santiago-Torres M, Mull KE, Sullivan BM, Ferketich AK, Bricker JB. Efficacy of an acceptance and commitment therapy-based smartphone application for helping rural populations quit smoking: Results from the iCanQuit randomized trial. *Prev Med*. 2022;157:10.700810.1016/j.ypmed.2022.107008.
14. Santiago-Torres M, Mull KE, Sullivan BM, Kendzor DE, Bricker JB. Efficacy and utilization of smartphone applications for smoking cessation among low-income adults: Secondary analysis of the iCanQuit randomized trial. *Drug Alcohol Depend*. 2022;231:109258.PMC8810613. 10.1016/j.drugalcdep.2021.109258.
15. Santiago-Torres M, Mull KE, Sullivan BM, Rigotti NA, Bricker JB. Acceptance and Commitment Therapy-Based Smartphone Applications for Cessation of Tobacco Use among Adults with High Nicotine Dependence: Results from the iCanQuit Randomized Trial. *Subst Use Misuse*. 2023:1–1110.10.80/10826084.2022.2161317.
16. Vilardaga R, Rizo J, Ries RK, Kientz JA, Ziedonis DM, Hernandez K, et al. Formative, multimethod case studies of learn to quit, an acceptance and commitment therapy smoking cessation app designed for people with serious mental illness. *Transl Behav Med*. 2019;9(6):1076–86.PMC7184916. 10.1093/tbm/iby097. [PubMed: 30445507]

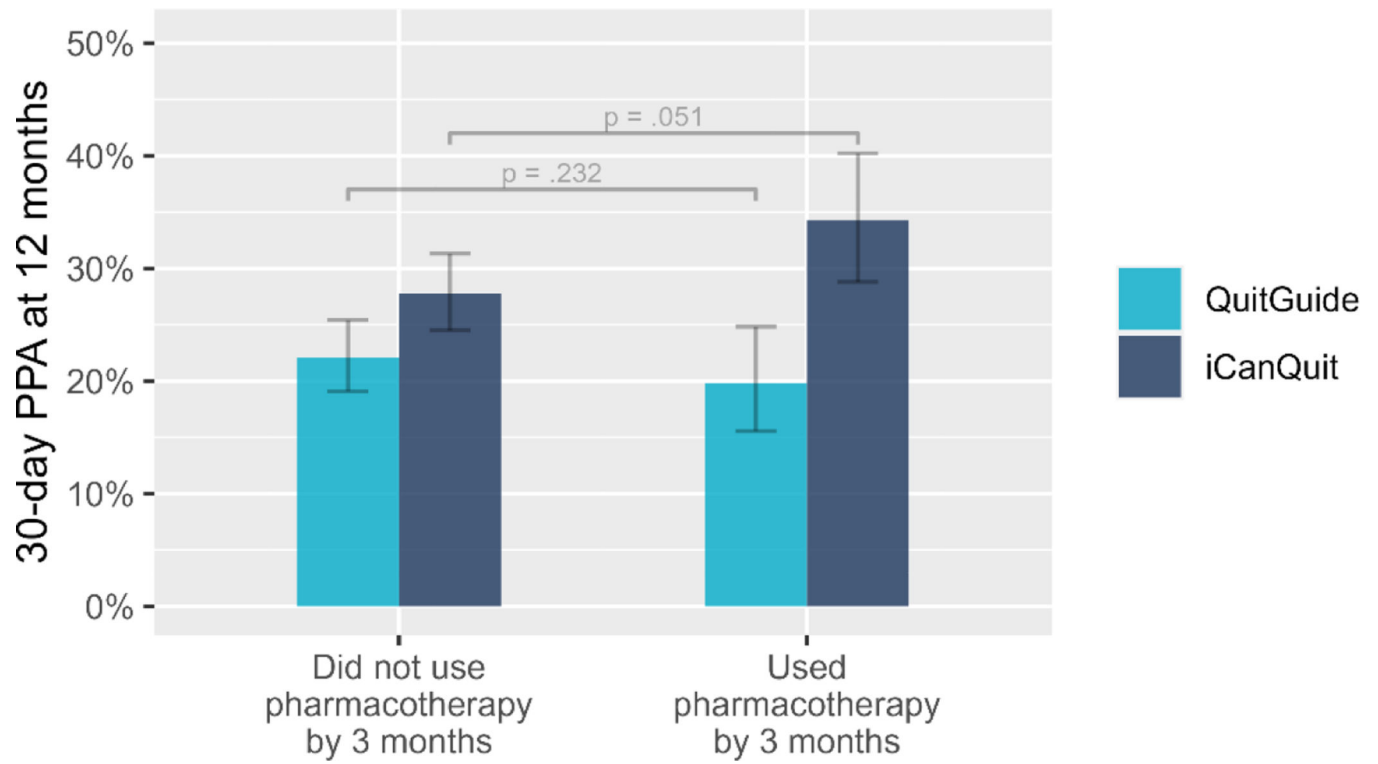
17. Fiore M, Jaén CR, Baker TB, Bailey WC, Bennett G, Benowitz NL, et al. A clinical practice guideline for treating tobacco use and dependence: 2008 update. A U.S. Public Health Service report. *Am J Prev Med.* 2008;35(2):158–76.PMC4465757. 10.1016/j.amepre.2008.04.009. [PubMed: 18617085]
18. Patnode CD, Henderson JT, Coppola EL, Melnikow J, Durbin S, Thomas RG. Interventions for Tobacco Cessation in Adults, Including Pregnant Persons: Updated Evidence Report and Systematic Review for the US Preventive Services Task Force. *JAMA.* 2021;325(3):280–9810.10.01/jama.2020.23541. [PubMed: 33464342]
19. Siu AL, Force USPST. Behavioral and Pharmacotherapy Interventions for Tobacco Smoking Cessation in Adults, Including Pregnant Women: U.S. Preventive Services Task Force Recommendation Statement. *Ann Intern Med.* 2015;163(8):622–3410.7326/M15–2023. [PubMed: 26389730]
20. Fan T, Blitz J. Behavioral and Pharmacotherapy Interventions for Tobacco Smoking Cessation in Adults, Including Pregnant Women. *Am Fam Physician.* 2016;93(10):861–2 [PubMed: 27175721]
21. Haddad A, Davis AM. Tobacco Smoking Cessation in Adults and Pregnant Women: Behavioral and Pharmacotherapy Interventions. *JAMA.* 2016;315(18):2011–210.01/jama.2016.2535. [PubMed: 27163990]
22. Hartmann-Boyce J, Hong B, Livingstone-Banks J, Wheat H, Fanshawe TR. Additional behavioural support as an adjunct to pharmacotherapy for smoking cessation. *Cochrane Database Syst Rev.* 2019;6(6):CD009670.PMC6549450. 10.1002/14651858.CD009670.pub4.
23. Baliunas D, Voci S, de Oliveira C, Selby P, Kurdyak P, Rosella L, et al. Association Between Smoking Cessation Treatment and Healthcare Costs in a Single-Payer Public Healthcare System. *Nicotine Tob Res.* 2023;25(1):86–93.PMC9717383. 10.1093/ntr/ntac166. [PubMed: 35792868]
24. Ekpu VU, Brown AK. The Economic Impact of Smoking and of Reducing Smoking Prevalence: Review of Evidence. *Tob Use Insights.* 2015;8:1–35.PMC4502793. 10.4137/TUI.S15628. [PubMed: 26242225]
25. Hollands GJ, Naughton F, Farley A, Lindson N, Aveyard P. Interventions to increase adherence to medications for tobacco dependence. *Cochrane Database Syst Rev.* 2019;8(8):CD009164.PMC6699660. 10.1002/14651858.CD009164.pub3.
26. Hartmann-Boyce J, Livingstone-Banks J, Ordonez-Mena JM, Fanshawe TR, Lindson N, Freeman SC, et al. Behavioural interventions for smoking cessation: an overview and network meta-analysis. *Cochrane Database Syst Rev.* 2021;1:CD01322910.02/14651858.CD013229.pub2.
27. Stead LF, Koilpillai P, Fanshawe TR, Lancaster T. Combined pharmacotherapy and behavioural interventions for smoking cessation. *Cochrane Database Syst Rev.* 2016;3(3):CD008286.PMC10042551. 10.1002/14651858.CD008286.pub3.
28. Kruse GR, Park ER, Chang Y, Haberer JE, Abrams LC, Shahid NN, et al. Proactively Offered Text Messages and Mailed Nicotine Replacement Therapy for Smokers in Primary Care Practices: A Pilot Randomized Trial. *Nicotine Tob Res.* 2020;22(9):1509–14.PMC7443591. 10.1093/ntr/ntaa050. [PubMed: 32198520]
29. Vidrine DJ, Frank-Pearce SG, Vidrine JI, Tahay PD, Marani SK, Chen S, et al. Efficacy of Mobile Phone-Delivered Smoking Cessation Interventions for Socioeconomically Disadvantaged Individuals: A Randomized Clinical Trial. *JAMA Intern Med.* 2019;179(2):167–74.PMC6439644. 10.1001/jamainternmed.2018.5713. [PubMed: 30556832]
30. Carrasco-Hernandez L, Jodar-Sanchez F, Nunez-Benjumea F, Moreno Conde J, Mesa Gonzalez M, Civit-Balcells A, et al. A Mobile Health Solution Complementing Psychopharmacology-Supported Smoking Cessation: Randomized Controlled Trial. *JMIR Mhealth Uhealth.* 2020;8(4):e17530.PMC7215523. 10.2196/17530.
31. Amiri S, Khan MAB. Digital interventions for smoking abstinence: a systematic review and meta-analysis of randomized control trials. *J Addict Dis.* 2023;41(1):4–2910.10.80/10550887.2022.2058300. [PubMed: 35426355]
32. Sha L, Yang X, Deng R, Wang W, Tao Y, Cao H, et al. Automated Digital Interventions and Smoking Cessation: Systematic Review and Meta-analysis Relating Efficiency to a Psychological Theory of Intervention Perspective. *J Med Internet Res.* 2022;24(11):e38206.PMC9713619. 10.2196/38206.

33. Gentili A, Failla G, Melnyk A, Puleo V, Tanna GLD, Ricciardi W, et al. The cost-effectiveness of digital health interventions: A systematic review of the literature. *Front Public Health*. 2022;10:787135.PMC9403754. 10.3389/fpubh.2022.787135.
34. Ben-Zeev D, Razzano LA, Pashka NJ, Levin CE. Cost of mHealth Versus Clinic-Based Care for Serious Mental Illness: Same Effects, Half the Price Tag. *Psychiatr Serv*. 2021;72(4):448–5110.1176/appi.ps.202000349. [PubMed: 33557599]
35. Ghani Z, Jarl J, Sanmartin Berglund J, Andersson M, Anderberg P. The Cost-Effectiveness of Mobile Health (mHealth) Interventions for Older Adults: Systematic Review. *Int J Environ Res Public Health*. 2020;17(15).PMC7432315. 10.3390/ijerph17155290.
36. Cunningham JA, Kushnir V, Selby P, Tyndale RF, Zawertailo L, Leatherdale ST. Beyond Quitting: Any Additional Impact of Mailing Free Nicotine Patches to Current Smokers? *Nicotine Tob Res*. 2018;20(5):654–5.PMC5892854. 10.1093/ntr/ntx106. [PubMed: 28521035]
37. Brown TJ, Todd A, O'Malley C, Moore HJ, Husband AK, Bambra C, et al. Community pharmacy-delivered interventions for public health priorities: a systematic review of interventions for alcohol reduction, smoking cessation and weight management, including meta-analysis for smoking cessation. *BMJ Open*. 2016;6(2):e009828.PMC4780058. 10.1136/bmjopen-2015-009828.
38. Cummings KM, Hyland A, Carlin-Menter S, Mahoney MC, Willett J, Juster HR. Costs of giving out free nicotine patches through a telephone quit line. *J Public Health Manag Pract*. 2011;17(3):E16–2310.97/PHH.0b013e3182113871. [PubMed: 21464679]
39. Cunningham JA, Kushnir V. Effectiveness of mass distribution of nicotine patches to promote tobacco control in rural versus urban settings. *J Epidemiol Community Health*. 2017;71(5):51910.1136/jech-2016–208103.
40. Fu SS, van Ryn M, Sherman SE, Burgess DJ, Noorbaloochi S, Clothier B, et al. Proactive tobacco treatment and population-level cessation: a pragmatic randomized clinical trial. *JAMA Intern Med*. 2014;174(5):671–710.10.01/jamainternmed.2014.177. [PubMed: 24615217]
41. Fu SS, van Ryn M, Nelson D, Burgess DJ, Thomas JL, Saul J, et al. Proactive tobacco treatment offering free nicotine replacement therapy and telephone counselling for socioeconomically disadvantaged smokers: a randomised clinical trial. *Thorax*. 2016;71(5):446–53.PMC4862067. 10.1136/thoraxjnl-2015-207904. [PubMed: 26931362]
42. Frieden TR. A framework for public health action: the health impact pyramid. *Am J Public Health*. 2010;100(4):590–5.PMC2836340. 10.2105/AJPH.2009.185652. [PubMed: 20167880]
43. Assmann SF, Pocock SJ, Enos LE, Kasten LE. Subgroup analysis and other (mis)uses of baseline data in clinical trials. *Lancet*. 2000;355(9209):1064–910.16/S0140-6736(00)02039-0. [PubMed: 10744093]
44. Radloff LS. The CES-D scale: a self-report depression scale for research in the general population. *Appl Psychol Meas*. 1977;1(3):385–40110.1177/014662167700100306.
45. Stein MB, Roy-Byrne PP, McQuaid JR, Laffaye C, Russo J, McCahill ME, et al. Development of a brief diagnostic screen for panic disorder in primary care. *Psychosom Med*. 1999;61(3):359–6410.97/00006842-199905000-00016. [PubMed: 10367617]
46. Lang AJ, Wilkins K, Roy-Byrne PP, Golinelli D, Chavira D, Sherbourne C, et al. Abbreviated PTSD Checklist (PCL) as a guide to clinical response. *Gen Hosp Psychiatry*. 2012;34(4):332–8.PMC3383936. 10.1016/j.genhosppsych.2012.02.003. [PubMed: 22460001]
47. Roy M, Dum M, Sobell LC, Sobell MB, Simco ER, Manor H, et al. Comparison of the quick drinking screen and the alcohol timeline followback with outpatient alcohol abusers. *Subst Use Misuse*. 2008;43(14):2116–2310.10.80/10826080802347586. [PubMed: 18825590]
48. Heatherton TF, Kozlowski LT, Frecker RC, Fagerstrom KO. The Fagerstrom Test for Nicotine Dependence: a revision of the Fagerstrom Tolerance Questionnaire. *Br J Addict*. 1991;86(9):1119–2710.1111/j.1360-0443.1991.tb01879.x. [PubMed: 1932883]
49. Graham JW. Missing data analysis: making it work in the real world. *Annu Rev Psychol*. 2009;60:549–7610.1146/annurev.psych.58.110405.085530. [PubMed: 18652544]
50. US Department of Agriculture, Economic Research Service, 2020. Rural-Urban Commuting Area Codes. Data accessed from <https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes/> on February 21, 2021. [Internet].
51. zipcode JB: U.S. ZIP Code database for geocoding. R package version 1.0. 2012.

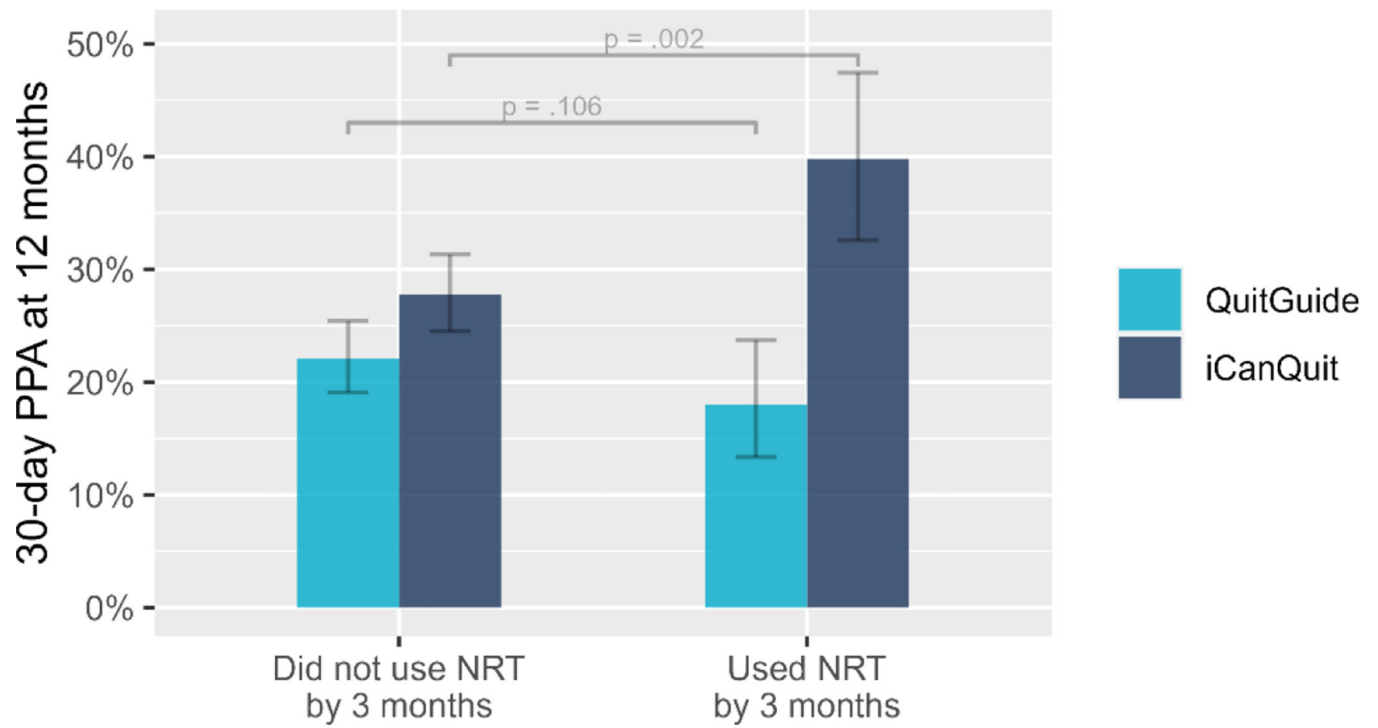


52. Chen X, Orom H, Hay JL, Waters EA, Schofield E, Li Y, et al. Differences in Rural and Urban Health Information Access and Use. *J Rural Health*. 2019;35(3):405–17.PMC6522336. 10.1111/jrh.12335. [PubMed: 30444935]
53. mice: multivariate imputation by chained equations in Stat Softw RJ. [Internet]. 2011.
54. Radloff LS. The CES-D Scale: a self-report depression scale for research in the general population. *Appl Psychol Meas*. 1977;1(3):385–401
55. Kernan WN, Viscoli CM, Makuch RW, Brass LM, Horwitz RI. Stratified randomization for clinical trials. *J Clin Epidemiol*. 1999;52(1):19–2610.10.16/s0895-4356(98)00138-3. [PubMed: 9973070]
56. Pocock SJ, McMurray JJV, Collier TJ. Statistical Controversies in Reporting of Clinical Trials: Part 2 of a 4-Part Series on Statistics for Clinical Trials. *J Am Coll Cardiol*. 2015;66(23):2648–6210.10.16/j.jacc.2015.10.023. [PubMed: 26670066]
57. Team RC. R: A language and environment for statistical computing. R Foundation for Statistical Computing. Vienna, Austria2022.
58. S.C. H, K.D. S, K.G. W. Acceptance and Commitment Therapy: The Process and Practice of Mindful Change. Second edition ed: The Guilford Press; 2016.
59. HS C, Get SS Out of Your Mind and Into Your Life: The New Acceptance and Commitment Therapy. Illustrated edition ed. New Harbinger Publications2005.
60. Hayes SC, Levin ME, Plumb-Villardaga J, Villatte JL, Pistorello J. Acceptance and commitment therapy and contextual behavioral science: examining the progress of a distinctive model of behavioral and cognitive therapy. *Behav Ther*. 2013;44(2):180–98.PMC3635495. 10.1016/j.beth.2009.08.002. [PubMed: 23611068]
61. Gifford EV, Kohlenberg BS, Hayes SC, Antonuccio DO, Piasecki MM, Rasmussen-Hall ML, et al. Acceptance-based treatment for smoking cessation. *Behavior Therapy*. 2004;35(4):689–705
62. Kushnir V, Sproule BA, Cunningham JA. Mailed distribution of free nicotine patches without behavioral support: Predictors of use and cessation. *Addict Behav*. 2017;67:73–810.10.16/j.addbeh.2016.12.008. [PubMed: 28039798]
63. Cunningham JA, Kushnir V, Selby P, Tyndale RF, Zawertailo L, Leatherdale ST. Effect of Mailing Nicotine Patches on Tobacco Cessation Among Adult Smokers: A Randomized Clinical Trial. *JAMA Intern Med*. 2016;176(2):184–9010.10.01/jamainternmed.2015.7792. [PubMed: 26809849]
64. Herbec A, Brown J, Shahab L, West R. Lessons learned from unsuccessful use of personal carbon monoxide monitors to remotely assess abstinence in a pragmatic trial of a smartphone stop smoking app - A secondary analysis. *Addict Behav Rep*. 2019;9:100122.PMC6542188. 10.1016/j.abrep.2018.07.003.
65. Thrul J, Meacham MC, Ramo DE. A novel and remote biochemical verification method of smoking abstinence: Predictors of participant compliance. *Tob Prev Cessat*. 2018;4.PMC6221470. 10.18332/tpc/90649.
66. van der Aalst CM, de Koning HJ. Biochemical verification of the self-reported smoking status of screened male smokers of the Dutch-Belgian randomized controlled lung cancer screening trial. *Lung Cancer*. 2016;94:96–10110.10.16/j.lungcan.2016.02.001. [PubMed: 26973213]
67. Wong SL, Shields M, Leatherdale S, Malaisson E, Hammond D. Assessment of validity of self-reported smoking status. *Health Rep*. 2012;23(1):47–53 [PubMed: 22590805]
68. Pineiro B, Vidrine DJ, Wetter DW, Hoover DS, Frank-Pearce SG, Nguyen N, et al. Implementation of Ask-Advise-Connect in a safety net healthcare system: quitline treatment engagement and smoking cessation outcomes. *Transl Behav Med*. 2020;10(1):163–7.PMC7295694. 10.1093/tbm/iby108. [PubMed: 30476236]
69. Patrick DL, Cheadle A, Thompson DC, Diehr P, Koepsell T, Kinne S. The validity of self-reported smoking: a review and meta-analysis. *Am J Public Health*. 1994;84(7):1086–93.PMC1614767. 10.2105/ajph.84.7.1086. [PubMed: 8017530]
70. Pierce JP, Gilpin EA. A minimum 6-month prolonged abstinence should be required for evaluating smoking cessation trials. *Nicotine Tob Res*. 2003;5(2):151–310.10.80/0955300031000083427. [PubMed: 12745486]
71. Hughes JR, Carpenter MJ, Naud S. Do point prevalence and prolonged abstinence measures produce similar results in smoking cessation studies? A systematic review. *Nicotine Tob Res*. 2010;12(7):756–62.PMC2893294. 10.1093/ntn/ntq078. [PubMed: 20504946]





**Figure 1.** Interaction effect of treatment arm and use of any FDA-approved pharmacotherapy on the primary 12-month smoking cessation outcome.



**Figure 2.** Interaction effect of treatment arm and use of FDA-approved NRT on the primary 12-month smoking cessation outcome.

Table 1.

## Major similarities and differences between iCanQuit and QuitGuide

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Major Similarities</b>                             | (1) Education and skills for preparing to quit and for preventing relapse after quitting including self-compassion, learning, and starting again; (2) Intention formation, including setting an actionable plan for quitting that includes setting a quit date; (3) Skills for coping with cravings to smoke; (4) Education on common triggers to smoke and barriers to cessation, nicotine withdrawal reactions, and how to seek support; (5) Step-by-step guide with content at 6 <sup>th</sup> grade or lower reading level. (6) Education on FDA-approved medications to aid cessation: (a) <i>advice to use</i> : recommendation that medications can help with feelings of withdrawal, advice that they can boost one's chances of quitting smoking; (b) <i>what are the FDA-approved medications</i> : describing what are nicotine gum, nicotine inhaler, nicotine lozenge, nicotine nasal spray, nicotine patch, bupropion pills, varenicline pills; (c) <i>where to obtain medications</i> : informed users that medications can be obtained over the counter at a pharmacy or grocery store or online (for all meds not requiring a prescription) or through a pharmacist (for all meds requiring a prescription); (d) <i>stating how these medicines work</i> (e.g., <i>reduce nicotine withdrawal</i> ); (e) <i>instructions on how to use</i> : neither app contained specific instructions on how to use medications but instead referred to user to read instructions that come with them and to talk to one's medical provider; (f) <i>common side effects</i> : briefly listed common side effects of medications (e.g., nausea). |                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Major Differences</b>                              | <b>iCanQuit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>QuitGuide</b>                                                                                                                                                                                                                                                                                                                                                                                               |
| <i>Approach to addressing motivation</i>              | (1) <b>Values</b> : Chosen life directions that guide goals and actions (e.g., major life areas, like family, that inspire you to be smoke-free); (2) <b>Testimonials</b> (e.g., 10–12 sentence audio-recorded stories from the program guide “Nancy”); (3) <b>Rewards</b> (e.g., earning visual “badges” of health progress contingent on number of smoke-free days)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (1) <b>Expectancies</b> : Beliefs about what actions will produce goal (e.g., listing expected outcomes of quitting smoking); (2) <b>Factual information processing</b> (e.g., listing ingredients of a cigarette); (3) <b>Risk perception</b> (e.g., risks of secondhand smoke and risks for smoking during pregnancy); (4) <b>Rewards</b> (e.g., showing health progress based on number of smoke-free days) |
| <i>Approach to addressing triggers to smoke</i>       | <b>Acceptance</b> : Change what you can and accept what you cannot. Openness to experience urges, emotions, and thoughts that trigger smoking (e.g., on-demand tips for letting urges come and go; progress tracking, experiential exercises on letting urges pass)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Avoidance</b> : Actively trying <u>not</u> to experience urges, emotions, and thoughts that trigger smoking (e.g., advice on avoiding triggers; advice on staying busy; recommendations for distracting yourself during an urge)                                                                                                                                                                            |
| <i>Approach to addressing relapse prevention</i>      | <b>Acceptance</b> : Perspective taking (e.g., writing a letter from your smoke-free future self); <b>Values</b> (e.g., making smoke-free vision statement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Avoidance</b> : Avoid high risk situations (e.g., avoid places where you used to smoke) and avoid urges (e.g., advice on how to fight cravings)                                                                                                                                                                                                                                                             |
| <i>Approach to addressing medication side effects</i> | Provided several tips on how to manage common side effects of each medication (e.g., for a skin rash, rotate site of the patch)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Provided general advice to use as directed to avoid common side effects or consult with medical provider                                                                                                                                                                                                                                                                                                       |

**Table 2.**

Baseline characteristics of participants reporting pharmacotherapy use at 3 months.

| Characteristic                         | <i>n</i> | Overall (n=2088) | Did not use<br>pharmacotherapy (n=1469) | Used pharmacotherapy<br>(n=619) | <i>P</i> value |
|----------------------------------------|----------|------------------|-----------------------------------------|---------------------------------|----------------|
| <i>Demographics</i>                    |          |                  |                                         |                                 |                |
| Age, mean (SD)                         | 2088     | 38.5 (10.9)      | 37.9 (10.9)                             | 39.8 (10.8)                     | <0.001         |
| Male                                   | 2088     | 606 (29%)        | 423 (29%)                               | 183 (30%)                       | 0.764          |
| Race                                   | 2088     |                  |                                         |                                 | 0.057          |
| White                                  |          | 1425 (68%)       | 981 (67%)                               | 444 (72%)                       |                |
| Black or African American              |          | 422 (20%)        | 308 (21%)                               | 114 (18%)                       |                |
| Asian                                  |          | 6 (<1%)          | 4 (<1%)                                 | 2 (<1%)                         |                |
| American Indian or Alaska Native       |          | 51 (2%)          | 35 (2%)                                 | 16 (3%)                         |                |
| Native Hawaiian or Pacific<br>Islander |          | 5 (<1%)          | 2 (<1%)                                 | 3 (<1%)                         |                |
| Multiple races                         |          | 150 (7%)         | 113 (8%)                                | 37 (6%)                         |                |
| Unknown                                |          | 29 (1%)          | 26 (2%)                                 | 3 (<1%)                         |                |
| Hispanic                               | 2088     | 173 (8%)         | 135 (9%)                                | 38 (6%)                         | 0.026          |
| Education                              | 2088     |                  |                                         |                                 | 0.258          |
| High school, GED or lower<br>education |          | 847 (41%)        | 613 (42%)                               | 234 (38%)                       |                |
| Some college, no college degree        |          | 790 (38%)        | 555 (38%)                               | 235 (38%)                       |                |
| College degree or higher               |          | 451 (22%)        | 301 (20%)                               | 150 (24%)                       |                |
| Employment                             | 2088     |                  |                                         |                                 | 0.008          |
| Employed                               |          | 1124 (54%)       | 804 (55%)                               | 320 (52%)                       |                |
| Unemployed                             |          | 262 (13%)        | 192 (13%)                               | 70 (11%)                        |                |
| Disabled                               |          | 310 (15%)        | 193 (13%)                               | 117 (19%)                       |                |
| Out of labor force                     |          | 392 (19%)        | 280 (19%)                               | 112 (18%)                       |                |
| Income                                 | 2088     |                  |                                         |                                 | 0.050          |
| <\$20,000/year                         |          | 752 (36%)        | 547 (37%)                               | 205 (33%)                       |                |
| \$20,000–\$4,499/year                  |          | 979 (47%)        | 688 (47%)                               | 291 (47%)                       |                |
| ≥\$55,000/year                         |          | 357 (17%)        | 234 (16%)                               | 123 (20%)                       |                |
| Urban residence                        | 2088     | 1610 (77%)       | 1129 (77%)                              | 481 (78%)                       | 0.715          |
| Married                                | 2088     | 665 (32%)        | 453 (31%)                               | 212 (34%)                       | 0.140          |
| LGBT                                   | 2088     | 349 (17%)        | 241 (16%)                               | 108 (17%)                       | 0.604          |
| <i>Mental Health</i>                   |          |                  |                                         |                                 |                |
| Depression positive screen             | 2078     | 988 (48%)        | 710 (49%)                               | 278 (45%)                       | 0.153          |
| Panic disorder positive screen         | 2055     | 567 (28%)        | 409 (28%)                               | 158 (26%)                       | 0.264          |
| PTSD positive screen                   | 2071     | 900 (43%)        | 647 (44%)                               | 253 (41%)                       | 0.211          |
| <i>Alcohol Use</i>                     |          |                  |                                         |                                 |                |
| Heavy drinker <sup>a</sup>             | 2026     | 288 (14%)        | 210 (15%)                               | 78 (13%)                        | 0.280          |
| <i>Smoking Behavior</i>                |          |                  |                                         |                                 |                |

| Characteristic                         | <i>n</i> | Overall (n=2088) | Did not use<br>pharmacotherapy (n=1469) | Used pharmacotherapy<br>(n=619) | <i>P</i> value |
|----------------------------------------|----------|------------------|-----------------------------------------|---------------------------------|----------------|
| FTCD score, mean (SD)                  | 2088     | 5.8 (2.1)        | 5.8 (2.1)                               | 6.0 (2.0)                       | 0.017          |
| Number of cigarettes/day, mean (SD)    | 2088     | 19.2 (14.6)      | 19.0 (14.7)                             | 19.6 (14.3)                     | 0.367          |
| Smokes more than one-half pack per day | 2088     | 1554 (74%)       | 1077 (73%)                              | 477 (77%)                       | 0.083          |
| First cigarette within 5 min of waking | 2088     | 1110 (53%)       | 765 (52%)                               | 345 (56%)                       | 0.138          |
| Smoked for ≥10 years                   | 2088     | 1749 (84%)       | 1210 (82%)                              | 539 (87%)                       | 0.009          |
| Used e-cigarettes in past month        | 2088     | 495 (24%)        | 354 (24%)                               | 141 (23%)                       | 0.554          |
| Quit attempts in past 12M, mean (SD)   | 1992     | 1.5 (5.9)        | 1.5 (6.5)                               | 1.5 (4.4)                       | 0.858          |
| Confidence to quit smoking, mean (SD)  | 2088     | 64.7 (26.8)      | 64.3 (27.0)                             | 65.8 (26.4)                     | 0.249          |
| Friend and partner smoking             |          |                  |                                         |                                 |                |
| Close friends who smoke, mean (SD)     | 2088     | 2.6 (1.7)        | 2.7 (1.7)                               | 2.6 (1.8)                       | 0.244          |
| Housemates who smoke, mean (SD)        | 2088     | 1.5 (0.9)        | 1.5 (0.9)                               | 1.4 (0.8)                       | 0.044          |
| Living with partner who smokes         | 2088     | 748 (36%)        | 543 (37%)                               | 205 (33%)                       | 0.104          |

<sup>a</sup>Heavy drinking is defined as 4 or more drinks on a typical drinking day for females and 5 or more drinks on a typical drinking day for males within the past 30 days.

**Table 3.**  
Association between any pharmacotherapy use and cessation outcomes in the iCanQuit and QuitGuide treatment arms.

| 12-month cessation outcome                            | Interaction of treatment and any pharmacotherapy use <sup>a</sup> , $\beta$ (95% CI), <i>P</i> value | Treatment arm | Overall                     | Did not use pharmacotherapy | Used pharmacotherapy       | OR (95% CI) <sup>b</sup> | <i>P</i> value |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------|-----------------------------|-----------------------------|----------------------------|--------------------------|----------------|
| 30-day PPA, complete case <sup>c</sup>                | 0.50 (0.24), <i>p</i> = 0.033                                                                        | iCanQuit      | 288 (30%), <i>n</i> =971    | 192 (28%), <i>n</i> =691    | 96 (34%), <i>n</i> =280    | 1.37 (1.00, 1.87)        | 0.051          |
| 30-day PPA, missing-as-smoking <sup>d</sup>           | 0.49 (0.23), <i>p</i> = 0.035                                                                        | QuitGuide     | 212 (21%), <i>n</i> =991    | 152 (22%), <i>n</i> =688    | 60 (20%), <i>n</i> =303    | 0.81 (0.57, 1.15)        | 0.232          |
| 30-day PPA, multiple imputation <sup>e</sup>          | 0.46 (0.23), <i>p</i> = 0.047                                                                        | iCanQuit      | 288 (28%), <i>n</i> =1041   | 192 (26%), <i>n</i> =741    | 96 (32%), <i>n</i> =300    | 1.35 (0.99, 1.83)        | 0.056          |
|                                                       |                                                                                                      | QuitGuide     | 212 (20%), <i>n</i> =1047   | 152 (21%), <i>n</i> =728    | 60 (19%), <i>n</i> =319    | 0.80 (0.56, 1.13)        | 0.198          |
|                                                       |                                                                                                      | iCanQuit      | 3076 (30%), <i>n</i> =10410 | 2057 (28%), <i>n</i> =7410  | 1019 (34%), <i>n</i> =3000 | 1.33 (0.97, 1.82)        | 0.079          |
|                                                       |                                                                                                      | QuitGuide     | 2219 (21%), <i>n</i> =10470 | 1587 (22%), <i>n</i> =7280  | 632 (20%), <i>n</i> =3190  | 0.82 (0.58, 1.15)        | 0.254          |
| 7-day PPA, complete case <sup>e</sup>                 | 0.42 (0.22), <i>p</i> = 0.060                                                                        | iCanQuit      | 347 (36%), <i>n</i> =971    | 239 (35%), <i>n</i> =691    | 108 (39%), <i>n</i> =280   | 1.20 (0.88, 1.62)        | 0.248          |
|                                                       |                                                                                                      | QuitGuide     | 287 (29%), <i>n</i> =991    | 207 (30%), <i>n</i> =688    | 80 (26%), <i>n</i> =303    | 0.78 (0.57, 1.07)        | 0.126          |
| 30-day PPA all nicotine/tobacco products <sup>f</sup> | 0.50 (0.25), <i>p</i> = 0.048                                                                        | iCanQuit      | 240 (25%), <i>n</i> =970    | 159 (23%), <i>n</i> =691    | 81 (29%), <i>n</i> =279    | 1.38 (1.00, 1.91)        | 0.052          |
|                                                       |                                                                                                      | QuitGuide     | 165 (17%), <i>n</i> =992    | 119 (17%), <i>n</i> =690    | 46 (15%), <i>n</i> =302    | 0.82 (0.56, 1.20)        | 0.306          |
| Prolonged abstinence: 3 and 12 months <sup>g</sup>    | 0.23 (0.36), <i>p</i> = 0.520                                                                        | iCanQuit      | 112 (14%), <i>n</i> =784    | 73 (13%), <i>n</i> =552     | 39 (17%), <i>n</i> =232    | 1.32 (0.85, 2.05)        | 0.215          |
|                                                       |                                                                                                      | QuitGuide     | 63 (8%), <i>n</i> =807      | 42 (8%), <i>n</i> =555      | 21 (8%), <i>n</i> =252     | 1.10 (0.63, 1.92)        | 0.732          |

<sup>a</sup>Interaction terms are provided on the log-odds scale. 'Any pharmacotherapy' may include NRT, varenicline and/or bupropion.

<sup>b</sup>All outcome models adjust for the four baseline factors used in stratified randomization: minority race or ethnicity, high school or less education, heavy smoking, and positive depression screen.

<sup>c</sup>Additional model covariates are gender, smoking for  $\geq 10$  years, confidence in being smokefree, living with a partner who smokes, and heavy drinking.

<sup>d</sup>Additional model covariates are gender, smoking for  $\geq 10$  years, confidence in being smokefree, number of close friends who smoke, and heavy drinking.

<sup>e</sup>Additional model covariates are age, gender, smoking for  $\geq 10$  years, confidence in being smokefree, living with a partner who smokes, and heavy drinking.

<sup>f</sup>Additional model covariates are gender, confidence in being smokefree, number of adults in home who smoke.

<sup>g</sup>Additional model covariates are positive PTSD screen, smoking for  $\geq 10$  years, using e-cigarettes, and confidence in being smokefree.



Table 4.

Association between use of NRT only and cessation outcomes in the iCanQuit and QuitGuide treatment arms.

| 12-month cessation outcome                            | Interaction of treatment and NRT use <sup>a</sup> , $\beta$ (95% CI), <i>p</i> value | Treatment arm | Overall                     | Did not use NRT             | Used NRT only              | OR (95% CI) <sup>b</sup> | <i>P</i> value |
|-------------------------------------------------------|--------------------------------------------------------------------------------------|---------------|-----------------------------|-----------------------------|----------------------------|--------------------------|----------------|
| 30-day PPA, complete case <sup>c</sup>                | 0.89 (0.27), <i>p</i> = 0.001                                                        | iCanQuit      | 262 (30%), <i>n</i> = 867   | 192 (28%), <i>n</i> = 691   | 70 (40%), <i>n</i> = 176   | 1.78 (1.24, 2.56)        | 0.002          |
|                                                       |                                                                                      | QuitGuide     | 193 (21%), <i>n</i> = 916   | 152 (22%), <i>n</i> = 688   | 41 (18%), <i>n</i> = 228   | 0.72 (0.48, 1.07)        | 0.106          |
| 30-day PPA, missing-as-smoking <sup>d</sup>           | 0.83 (0.27), <i>p</i> = 0.002                                                        | iCanQuit      | 262 (28%), <i>n</i> = 934   | 192 (26%), <i>n</i> = 741   | 70 (36%), <i>n</i> = 193   | 1.65 (1.16, 2.34)        | 0.005          |
|                                                       |                                                                                      | QuitGuide     | 193 (20%), <i>n</i> = 968   | 152 (21%), <i>n</i> = 728   | 41 (17%), <i>n</i> = 240   | 0.71 (0.48, 1.06)        | 0.091          |
| 30-day PPA, multiple imputation <sup>e</sup>          | 0.80 (0.27), <i>p</i> = 0.003                                                        | iCanQuit      | 2803 (30%), <i>n</i> = 9340 | 2057 (28%), <i>n</i> = 7410 | 746 (39%), <i>n</i> = 1930 | 1.66 (1.15, 2.39)        | 0.007          |
|                                                       |                                                                                      | QuitGuide     | 2023 (21%), <i>n</i> = 9680 | 1587 (22%), <i>n</i> = 7280 | 436 (18%), <i>n</i> = 2400 | 0.74 (0.50, 1.09)        | 0.125          |
| 7-day PPA, complete case <sup>e</sup>                 | 0.64 (0.25), <i>p</i> = 0.011                                                        | iCanQuit      | 315 (36%), <i>n</i> = 867   | 239 (35%), <i>n</i> = 691   | 76 (43%), <i>n</i> = 176   | 1.47 (1.03, 2.11)        | 0.032          |
|                                                       |                                                                                      | QuitGuide     | 267 (29%), <i>n</i> = 916   | 207 (30%), <i>n</i> = 688   | 60 (26%), <i>n</i> = 228   | 0.77 (0.54, 1.10)        | 0.153          |
| 30-day PPA all nicotine/tobacco products <sup>f</sup> | 0.82 (0.29), <i>p</i> = 0.004                                                        | iCanQuit      | 219 (25%), <i>n</i> = 866   | 159 (23%), <i>n</i> = 691   | 60 (34%), <i>n</i> = 175   | 1.77 (1.22, 2.58)        | 0.003          |
|                                                       |                                                                                      | QuitGuide     | 152 (17%), <i>n</i> = 917   | 119 (17%), <i>n</i> = 690   | 33 (15%), <i>n</i> = 227   | 0.77 (0.50, 1.18)        | 0.231          |
| Prolonged abstinence: 3 and 12 months <sup>g</sup>    | 0.45 (0.42), <i>p</i> = 0.288                                                        | iCanQuit      | 98 (14%), <i>n</i> = 699    | 73 (13%), <i>n</i> = 552    | 25 (17%), <i>n</i> = 147   | 1.39 (0.83, 2.33)        | 0.205          |
|                                                       |                                                                                      | QuitGuide     | 55 (7%), <i>n</i> = 742     | 42 (8%), <i>n</i> = 555     | 13 (7%), <i>n</i> = 187    | 0.91 (0.47, 1.74)        | 0.769          |

<sup>a</sup>Interaction terms are provided on the log-odds scale.<sup>b</sup>All outcome models adjust for the four baseline factors used in stratified randomization: minority race or ethnicity, high school or less education, heavy smoking, and positive depression screen.<sup>c</sup>Additional model covariates are gender, smoking for  $\geq 10$  years, confidence in being smokefree, living with a partner who smokes, and heavy drinking.<sup>d</sup>Additional model covariates are gender, smoking for  $\geq 10$  years, confidence in being smokefree, number of close friends who smoke, and heavy drinking.<sup>e</sup>Additional model covariates are age, gender, smoking for  $\geq 10$  years, confidence in being smokefree, living with a partner who smokes, and heavy drinking.<sup>f</sup>Additional model covariates are gender, confidence in being smokefree, number of adults in home who smoke.<sup>g</sup>Additional model covariates are positive PTSD screen, smoking for  $\geq 10$  years, using e-cigarettes, and confidence in being smokefree.